

Unlocking the Mysteries of the Ice Ages

Connecting Orbital Theory to the “41,000-Year World”

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Caltech - ESE 101: Earth's Atmosphere

- **The Standard Model:** Milanković Theory and the orbital mechanics learned in class.
- **The Puzzle:** The “41,000-Year World” paradox and the missing Precession signal.
- **The Physical Solutions:**
 - ▶ **Mechanism 1:** Integrated Summer Insolation (Kepler’s Cancellation).
 - ▶ **Mechanism 2:** Anti-phased Hemispheres (Signal Cancellation).
- **The Transition:** Explaining the shift to 100kyr cycles (Regolith Hypothesis).
- **Future Implications:** Modern ice sheet stability and the end of orbital pacing in the Anthropocene.

The Standard Model: Orbital Forcing Mechanics

Orbital Forcing: As discussed in class, Earth's orbit varies in three primary ways:

1. **Eccentricity (100kyr/405kyr):** The shape of the orbit. Weak total annual energy forcing ($\sim 10^{-3}$).
2. **Obliquity (41kyr):** The tilt of the axis. Directly controls the severity of seasonality.
3. **Precession (19-24kyr):** The wobble of the axis. Determines *where* in the orbit the solstices/equinoxes occur.

(Source: Schneider Ch. 3)

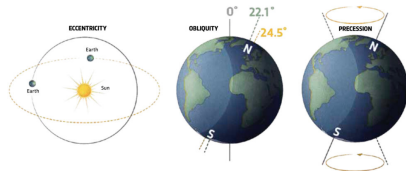


Figure 1: The three primary orbital parameters that modulate insolation over long timescales.

The Standard Model: Milanković's Theory

- **Milanković's 1941 Hypothesis:** Ice ages are driven by a single, critical factor: **peak summer insolation at 65°N** (Milanković, 1941).
- **The Rationale:** The growth of continental ice sheets requires cool summers (to prevent melt), not necessarily cold winters.
- **The Mechanism:** Weak summer intensity allows snow/ice from the previous winter to survive and accumulate.

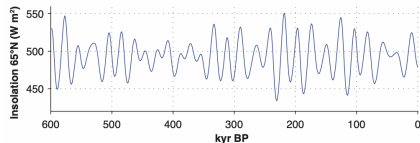


Figure 2: Insolation at 65°N at northern summer solstice. Milankovitch argued that glaciations begin when this insolation is weak (Figure 3.13, Schneider Ch. 3).

The Prediction vs. Reality

- **Prediction:** Precession (23kyr) dominates summer insolation (Panel c).
- **Reality:** Geologic $\delta^{18}O$ records (Panel a) show **only** 41kyr cycles.
- **The Paradox:** Where is the Precession signal?

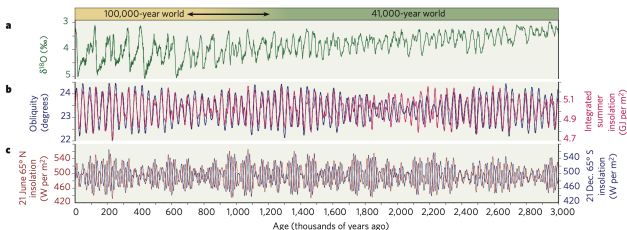


Figure 3: The “41,000-Year World” (Raymo & Huybers, 2008).

Solution 1: Integrated Summer Insolation

- **Hypothesis:** Ice sheet melting depends on the **total energy** received over the summer (degree days), not just the peak intensity (Huybers, 2006).
- **Kepler's Second Law:** As derived in class, Earth moves faster when closer to the Sun (Perihelion).
 - ▶ High Intensity → Shorter Duration.
 - ▶ Low Intensity → Longer Duration.
- **The Result:** When integrating energy over the full summer, intensity and duration cancel out.
- **Conclusion:** Integrated Summer Insolation is **independent of precession**, leaving Obliquity (41kyr) as the dominant driver (Raymo & Huybers, 2008).

The Mechanism: Kepler's Cancellation

We can prove why the Precession signal disappears using the integral derived in class (Eq 3.14, Schneider Ch. 3).

- **The Intensity:** Solar Flux S follows the inverse square law ($S \propto 1/d^2$).
- **The Duration:** By Kepler's 2nd Law, the time spent in an orbital arc dt is proportional to d^2 (conservation of angular momentum).

The Cancellation

When we integrate the total energy J over a season:

$$J = \int_{t_1}^{t_2} S dt \propto \int_{\lambda_1}^{\lambda_2} \left(\frac{1}{d^2} \right) \cdot (d^2) d\lambda$$

The d^2 terms cancel out exactly!

- **Result:** The total energy depends only on Obliquity and the length of the season (eccentricity), but **not** on the longitude of perihelion (Precession).

Solution 2: Anti-Phased Hemispheres

- **Hypothesis:** Northern and Southern ice sheets were both dynamic but out of phase (Raymo et al., 2006).
- **Orbital Physics:**
 - ▶ **Precession:** Anti-phased between hemispheres. (A bright Northern summer = A dim Southern summer).
 - ▶ **Obliquity:** In-phase globally. (High tilt = strong seasonality everywhere).
- **The Signal Cancellation:** Marine $\delta^{18}O$ records measure *global* ice volume.
- If the North grew while the South shrank (due to precession), the net change is near zero, hiding the precession signal in global records.

The Mechanism: Hemispheric Cancellation

- We can model the global $\delta^{18}O$ signal as the sum of Northern and Southern ice volume.
- **Precession (P):** Out of phase.

$$P_{North} \approx \sin(\omega t) \quad \text{and} \quad P_{South} \approx \sin(\omega t + \pi) = -\sin(\omega t)$$

- **Obliquity (ϵ):** In phase.

$$\epsilon_{North} \approx \sin(\Omega t) \quad \text{and} \quad \epsilon_{South} \approx \sin(\Omega t)$$

The Global Signal ($\delta^{18}O$)

The ocean mixes the signal from both poles:

$$\Delta \text{Global Ice} = \Delta V_{North} + \Delta V_{South}$$

- **Precession Component:** $\sin(\omega t) + (-\sin(\omega t)) \approx 0$ (Cancellation!)
- **Obliquity Component:** $\sin(\Omega t) + \sin(\Omega t) \approx 2 \sin(\Omega t)$ (Amplification!)

The 100kyr Transition: The Regolith Hypothesis

- The paper addresses the shift from the 41k world to the 100k cycles of the last 1 million years.
- **Regolith Hypothesis (Clark & Pollard, 1998):**
 - ▶ **Early Pleistocene:** Ice sheets sat on loose, slippery sediment (regolith). Thin, mobile, and responsive to 41k cycles.
 - ▶ **The Change:** Repeated glaciations eroded the regolith down to hard bedrock.
 - ▶ **Late Pleistocene:** Ice sheets stuck to bedrock (high friction). Thicker, stable, allowing 100kyr growth.

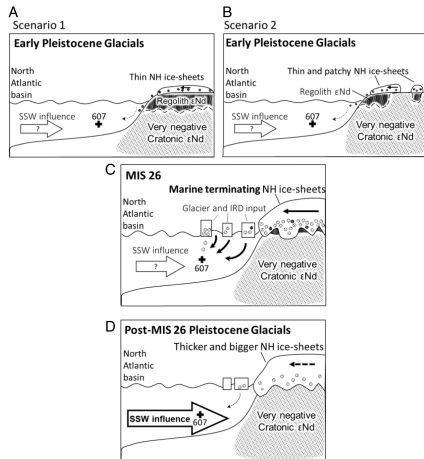


Figure 4: Erosion of slippery regolith leads to thicker, stickier ice sheets (Clark & Pollard, 1998).

Future Implication: The End of Orbital Pacing?

- **The Historical Rule:** Raymo & Huybers (2008) showed how subtle orbital nudges ($< 5 \text{ W/m}^2$) paced the growth and decay of ice sheets for the last 3 million years.
- **The Anthropocene Rupture:** Modern modeling (Ganopolski et al., 2016) identifies a “critical CO_2 threshold” for initiating a new ice age.
 - ▶ Even with orbital conditions favoring glaciation (low obliquity), nucleation fails if $\text{CO}_2 > 280 \text{ ppm}$.
- **Current Reality ($\sim 420 \text{ ppm}$):** We have largely overridden the orbital signal.

Conclusion

The “41,000-Year World” and “100,000-Year World” dynamics are likely suspended. We have entered a no-analog state where anthropogenic forcing dominates the delicate orbital mechanisms described in the paper.

Greenland Ice Sheet & Paleoclimate Connection

GrIS Evolution, 24 ka–Present

- Data-constrained model of GrIS since LGM.
- Rapid retreat during high summer energy periods.
- Strong feedbacks:
 - ▶ albedo loss
 - ▶ ocean–ice interaction
 - ▶ basal meltwater flow
- Nonlinear response: small climate shifts → large ice loss.

Connection to Article

- Past ice sheets controlled by **summer insolation thresholds** → Greenland shows same threshold behavior.
- Both emphasize **nonlinear dynamics** (feedbacks, tipping points).
- Mid-Pleistocene sensitivity shift parallels modern GrIS:
 - ▶ more basal meltwater
 - ▶ faster ice flow
 - ▶ grounding-line retreat
- Past deglaciation patterns help project future GrIS instability + sea-level risk.

- **Beyond the State of the Art:**

- ▶ Raymo & Huybers (2008) challenged the rigid “Peak Insolation” view.
- ▶ Demonstrated that **duration** (integrated energy) is as critical as intensity for ice melt.
- ▶ Revealed how hemispheric asymmetry can mask strong local orbital forcing in global records.

- **Future Implications:**

- ▶ **Greenland Vulnerability:** Modern ice sheets face the same nonlinear threshold behaviors described in the Early Pleistocene (e.g., albedo feedbacks, basal lubrication).
- ▶ **The Anthropocene Rupture:** Current CO₂ levels (~ 420 ppm) have likely overridden the natural orbital pacing mechanism, locking Earth out of the next glacial inception.

References

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-  Clark, P. U., & Pollard, D. (1998). Origin of the middle Pleistocene transition by ice sheet erosion of regolith. *Paleoceanography*, 13(1), 1-9.
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-  Barker, S., et al. (2011). 800,000 Years of Abrupt Climate Variability. *Science*, 334(6054), 347-351.
-  Briner, J. P., et al. (2025). The Greenland Ice Sheet evolution over the last 24,000 years: insights from model simulations evaluated against ice-extent markers. *The Cryosphere*, 19, 5719-5745.
-  ESE 101 Class Materials: *Climate Chapter 3: Earth's Orbit and Insolation*.

Thank You!

Questions?

APPENDIX: Obliquity and Annually Averaged Insolation

- **Latitudinal Distribution:** Annually averaged insolation is currently maximal at the equator because Earth's obliquity γ is 23.4° .
- **The Critical Threshold:** Annual-mean insolation would be maximal at the poles only if obliquity exceeded a critical value of $\approx 54^\circ$.
- **Comparison to Uranus:** With an obliquity of 97.86° , Uranus orbits “on its side,” resulting in an annually averaged maximum at the poles.
- **Significance for Ice Ages:** Earth's relatively low obliquity ensures that polar regions receive significantly less energy than lower latitudes, making high-latitude summer insolation the primary limiting factor for ice persistence.

The Seasonal Trigger

Obliquity variations modulate the **seasonal contrast**. Low tilt minimizes summer melt capacity at 65°N , providing the orbital “pacing” for ice sheet inception.

APPENDIX: Internal Climate Feedback Amplification

- **Mechanism of Amplification:** Weak orbital pacing ($3 - 5 \text{ W/m}^2$) is transformed into massive glacial transitions via internal carbon cycle and albedo feedbacks.
- **Surface Temperature Modeling (T_s):** Modeled via Radiative-Convective Equilibrium (RCE):

$$T_s = T_e + \Gamma H_e$$

where T_e is the emission temperature, Γ is the lapse rate, and H_e is the height of emission.

- **CO₂ Forcing:** Reduced glacial CO₂ concentrations effectively lower the emission height (H_e) into a warmer, deeper part of the atmosphere.
- **Thermal Adjustment:** To maintain the forced tropospheric lapse rate, the entire profile shifts to colder temperatures, resulting in the large global temperature anomalies observed in the geologic record.

APPENDIX: Peak Daily vs. Integrated Summer Insolation

- **Peak Daily Insolation:** Dominated by the 23kyr Precession signal. Varies significantly based on the Earth-Sun distance at the solstice.
- **Integrated Flux Hypothesis:** Ice melting corresponds to the total seasonal energy (degree days).
- **Flux Integration Logic:**

$$J = \int S dt \propto \int \frac{1}{d^2} \cdot d^2 d\lambda$$

- The integration cancels the distance squared (d^2) factor introduced by Kepler's 2nd Law.

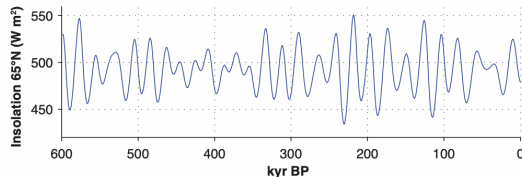


Figure 5: Calculated time series showing the disappearance of the 23kyr signal in integrated flux metrics (Schneider Ch. 3).

APPENDIX 3: Spectral Proof of the "41k World"

- **Power Spectral Density (PSD):** Analysis of the $\delta^{18}O$ record decomposes climate variability into periodic components.
- **The Evidence (Panel a):** For the past 3 million years, the spectrum shows a dominant peak at **41 kyr** (Obliquity).
- **The "Missing" Signal:** Note the absence of a distinct peak at **23 kyr** (Precession) in the Early Pleistocene record, despite Precession dominating the local insolation forcing.

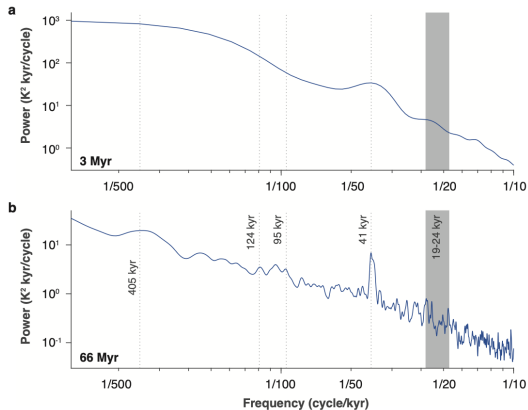


Figure 6: Power spectrum of temperature reconstructions. Note the dominant 41kyr peak (blue line) in the 3 Myr record.

APPENDIX 4: The Standard Insolation Model (Eq 3.20)

- **Milanković's Metric:** The standard theory relies on *Peak Daily Insolation* or *Diurnally Averaged Insolation* ($\overline{\cos \theta}$).
- **The Equation:**

$$\overline{\cos \theta} = \frac{1}{\pi} (\eta_d \sin \phi \sin \delta + \cos \phi \cos \delta \sin \eta_d)$$

- **Why Precession Dominates Here:** When calculating the total flux F , this term is multiplied by the distance factor $(d_0/d)^2$:

$$F = S_0 \left(\frac{d_0}{d} \right)^2 \overline{\cos \theta}$$

- Because d (Earth-Sun distance) varies with Precession (23kyr), this metric introduces the strong 23kyr signal that contradicts the geologic record.